

Independent random variables

Two RV's X and Y are independent if and only if one of the followings holds:

$$\bullet \quad \begin{cases} P_{X,Y}(x,y) = P_X(x) \cdot P_Y(y), \quad \forall x,y \\ f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y), \quad \forall x,y \end{cases}$$

$$\bullet \quad F_{X,Y}(x,y) = F_X(x) \cdot F_Y(y), \quad \forall x,y$$

Notation : $X \perp\!\!\!\perp Y$

Ex. Let X and Y have joint

PDF :

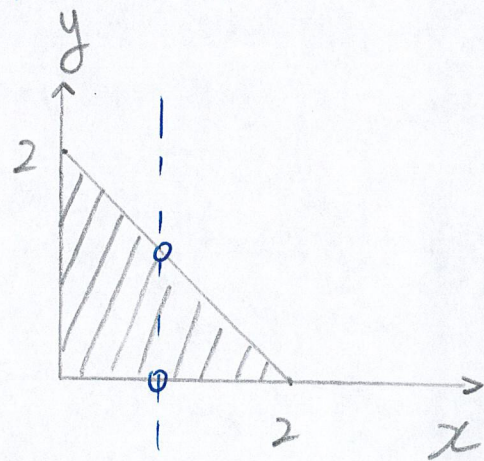
$$f_{X,Y}(x,y) = \begin{cases} \frac{3}{2} x \cdot y, & x \geq 0, y \geq 0 \\ & x+y \leq 2 \\ 0, & \text{o.w.} \end{cases}$$

Are X and Y independent?

$$f_X(x) = \int_0^{2-x} \frac{3}{2} x \cdot y \, dy$$

$$= \frac{3}{2} x \cdot \frac{1}{2} y^2 \Big|_0^{2-x}$$

$$= \frac{3}{4} x \cdot (2-x)^2, \quad 0 \leq x \leq 2$$



$$\therefore f_{X,Y}(x,y) = f_{X,Y}(y,x), \quad \forall x,y$$

In other words, X and Y are
exchangeable.

$$\therefore f_Y(y) = \frac{3}{4} \cdot y \cdot (2-y)^2, \quad 0 \leq y \leq 2$$

When $(x,y) = (1,1)$, then

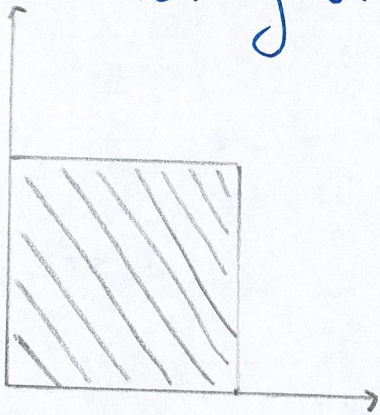
$$\bullet f_{X,Y}(1,1) = \frac{3}{2} \cdot 1 \cdot 1 = \frac{3}{2}$$

$$\bullet f_X(1) \cdot f_Y(1) = \left[\frac{3}{4} \cdot 1 \cdot (2-1)^2 \right]^2 = \left(\frac{3}{4} \right)^2$$

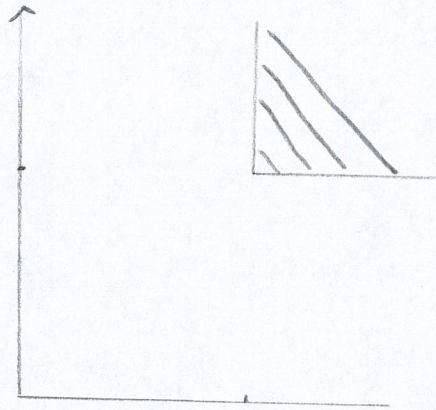
$$\therefore f_{X,Y}(1,1) \neq f_X(1) \cdot f_Y(1)$$

$$\therefore X \not\sim Y$$

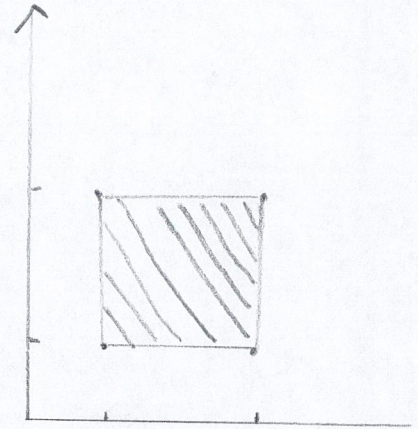
NOTE : For continuous RV's to be independent, the supports must be rectangles.



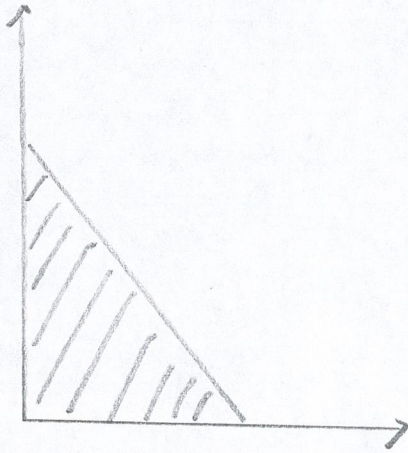
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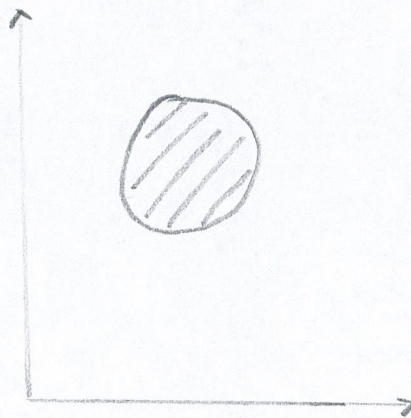
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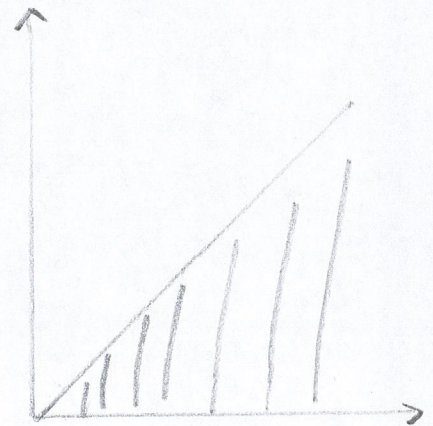
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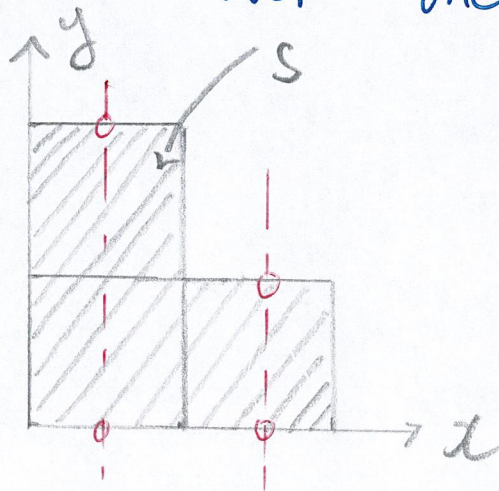
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★ NOTE: Rectangle support is a necessary but NOT sufficient condition, i.e.,

rectangle support $\Rightarrow$ maybe independent

non-rectangle support $\Rightarrow$ must be dependent

III Ex. RV's (X, Y) have a constant joint PDF over the following area:



Are X and Y independent?

$$\iint_A f_{x,y}(x,y) dx dy$$

= volume above S

$$= 1$$

$$C = \frac{1}{\text{area of } S} = \frac{1}{3}$$

$$f_{x,y}(x,y) = \begin{cases} \frac{1}{3}, & (x,y) \in S \\ 0, & \text{o.w.} \end{cases}$$

Moreover,

$$f_x(x) = \begin{cases} \int_0^2 \frac{1}{3} dy = \frac{2}{3}, & x \in (0,1) \\ \int_0^1 \frac{1}{3} dy = \frac{1}{3}, & x \in [1,2) \end{cases}$$

∴ X and Y are exchangeable.

$$\therefore f_Y(y) = \begin{cases} \frac{2}{3} & , y \in (0, 1) \\ \frac{1}{3} & , y \in [1, 2) \end{cases}$$

When $x = 1.5$ and $y = 0.5$

$$\frac{1}{3} = f_{X,Y}(x, y)$$

$\neq$

$$f_X(x) \cdot f_Y(y) = \frac{2}{9}$$

∴ $X \not\sim Y$.